

Functional Data Processing Pipeline with Apache Spark

Final Project in Functional Programming

PLEASE NOTE THAT DURING THE MEETINGS TILL THE END OF THE SEMESTER CHANGES MIGHT BE INTRODUCED INTO THIS DOCUMENT. **THERE WON'T BE ANY CHANGE IN THE REQUIREMENTS.** THE ONLY CHANGE POSSIBLE ARE FOR IMPROVING THE DOCUMENT ITSELF (SIMPLIFY ITS UNDERSTANDING). THEREFORE, MAKE SURE THAT YOUR PROJECT MEETS THE REQUIREMENTS OF THIS DOCUMENT BY THE END OF THIS SEMESTER

THE FASTEST WAY POSSIBLE FOR GETTING CLARIFICATIONS REGARDING THIS DOCUMENT IS TO WRITE EACH QUESTION AS A SEPARATE POST IN OUR COURSE'S FORUM.

QUESTIONS REGARDING THE INTERPRETATION OF THIS DOCUMENT SHOULD BE POSTED IN THE COURSE FORUM. MAKE SURE TO FOLLOW THIS FORUM TO VERIFY THAT YOUR INTERPRETATION IS THE ACCURATE ONE.

Introduction

Students will design and implement a complete data analytics pipeline using **Apache Spark** and **Scala**, applying **functional programming principles** such as immutability, pure functions, higher-order functions, currying, and combinators. The pipeline should process and analyze real-world-like data (e.g., CSV or JSON), and provide meaningful outputs in a functional and scalable manner.

Learning Goals

The goals to achieve in this project are the following:

- Apply **pure functions** and avoid mutable state
- Leverage **higher-order functions**, **currying**, and **closures**
- Use **pattern matching**, **case classes**, and **traits**

- Employ **Spark transformations** (map, flatMap, filter, reduceByKey, etc.)
- Compose functions using **combinators**
- Structure code using **modular, functional design**

Project Requirements

Data Source

Choose or generate a dataset with at least 10,000 records. Examples:

- Public domain movie data
- GitHub activity logs
- COVID-19 statistics
- E-commerce transactions

Functional Core

Implement all transformations (e.g., filtering, grouping, summarizing) as pure functions. Use immutable data structures (e.g., case classes, List, Map) when possible. Use currying or partial functions where applicable. Demonstrate function composition when applicable.

Apache Spark Usage

Use the RDD API or the DataFrame API or the DataSet API. You should use at least **four** different Spark operations (e.g., map, filter, reduceByKey, groupBy, join). You should load data from an external source and save the results to a file.

Advanced Functional Programming

You should include and document at least **three** of the following:

A custom **combinator**

Use of **closures** in Spark transformations

A **tail-recursive** function for computation

Pattern matching with case classes

Functional error handling

Documentation & Testing

Inline comments and ScalaDoc should be added. Unit testing should be added using ScalaTest. You should separate the pure logic from the I/O operations. The separation should be clear.

Code Style

The code in Scala should follow the style guide at <https://tinyurl.com/javapoints> with the following changes. If you have questions regarding this style guide, please post them in the course forum.

Change 1: In the Comments section, instead of using JavaDoc you should use ScalaDoc.

Change 2: In the Identifiers section, instead of “Class/Enum/Exception/Interface” it should be “Class/Object/Exception/Trait”.

Change 3: In the Classes section, ignore item 9, and in item 2, make sure you define a property (not a setter or a getter)

Change 4: The interfaces section applies to traits.

Change 5: Please ignore the Exceptions Handling, Primitive Types, IOStreams, Enum, and Threads sections.

Submission Guidelines

Here are the guidelines for submitting this project. You should carefully follow these guidelines. If a question arises, you should post it to the course forum to get a detailed, accurate answer. Points will be deducted when the submission doesn't meet the following guidelines (e.g., if the submitted PDF is not adequately organized to allow code review, points will be deducted).

0. You should develop the project with the IntelliJ IDE. It can be the ultimate version or the community one. Both of them are OK. You should use JDK 11, and Scala 2.12.19.

1. You should create a short video (try to make it up to 60s... if you want a longer video, then make a longer video) that shows how the project runs. You should upload that video to YouTube and make sure you upload it as an **unlisted** video. The video should include your explanations for advanced functional programming techniques implemented in your project and the Spark operations you were using. The

explanation should be in your voice. It cannot be in the text that your video shows.

2. You should pack the entire project into a **ZIP** file (You should choose in your IntelliJ IDE File->Export->Project to Zip file), and together with the **PDF** file, you should upload these two files to the submission box (it will be opened on Moodle). The names of the ZIP and the PDF files should be the first name + “_” + the last name (in English) of the development team manager (e.g. if the team manager is Moshe Israeli then the name of the files should be moshe_israeli.zip and moshe_israeli.pdf). The names of the files you submit cannot include spaces, and cannot include special characters. They should include English letters only. You can use an underscore as a replacement for a simple space.

3. You should create a PDF file (one single PDF file!!!) and copy to that file all code files that were coded by you. Make sure that lines are not broken. Make sure the alignment is to the left. Make sure this PDF file is properly organized to allow the code review.

4. The PDF file should include (at the beginning of it) the following:

- a. The first name and the last name of the development team manager.
- b. First name + Last name + ID + Mobile Number + Email Address of each one of the team members.
- c. Link to the video you created (item 1). The link should be clickable.
- d. Short explanation for the Spark operations and the advanced functional programming techniques you implemented in your project. The explanation should include the names of the classes and functions where the advanced functional programming techniques were implemented.
- e. The slides of the presentation you presented.
- f. The PDF should include a summary (up to 100 words) of your use of the collaboration tools that were presented in class (at least 2 of them).

5. The team manager should submit the two files (ZIP+PDF) in the assignment box that will be opened in our course on moodle (Only The Team Manager Should Submit The Project!!!! The Other Students Don't Need To Submit!). Please note the time difference between the time on the server (on which Moodle is running) and the time on your end. Due to this difference, you should treat the deadline published on our Moodle website as if it was 30min earlier. Late submissions won't be accepted. Submissions of projects developed by a single student won't be accepted. It is not possible to get a delay. Teams with justified reasons for getting a delay (in accordance with the college guidelines) will be handled separately. They won't submit through the assignment box of this project, and they won't get their mark with all others.

6. The deadline for submitting this project will be published in the course message board (Moodle).

7. The project will be carried out in teams of two students and will also include a 10% teamwork assessment using the tools presented during the course. A student who submits the project alone will lose 10% of the project grade. Students who submit alone will automatically lose 10% from the project grade.

8. When submitting a project it is not possible to get a delay. Students that don't submit on time will lose 2pts for every hour of delay.

Questions and Answers

1.

If we use the Function1 trait's function `compose` and `andThen` will that be considered to be an implementation of the Combinator design pattern in Scala.

=> Yes. Please note that doing so won't be considered a custom combinator implementation.

2.

When presenting the project in class, we should create a presentation that its slides that are added to a PDF document. What should that presentation include?

=> This presentation should describe the project you developed, the findings and the functional programming principles that were implemented.

Document Changes